MILITARY SPECIFICATION

MONITOR MODULE, STATUS, ID-1921/FSS-9(V)

AND

MONITOR MODULE, ALARM, ID-1922/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the two monitor modules of the Joint-Services Interior Intrusion Detection System (J-SIIDS) which are used to display the alarm and status of each protected zone to the monitor operator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

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MIL-P-116	- Preservation, Methods of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-C-52853	- Control Unit, Alarm Set: C-9412/FSS-9(V)
MIL-C-52876	- Cabinets, Monitor: CY-7359/FSS-9(V), CY-7360/FSS-9(V). and CY-7361/FSS-9(V).
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810[1/]	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration System Requirements.

1/ The "C" revision of MIL-STD-810 shall be used.

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2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

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TA13220E3950	- Monitor Module, Status - ID-1921/FSS-9(V).
TA13220E4050	- Monitor Module, Alarm - ID-1922/FSS-9(V).
TA13220E3952	- Printed Wiring Board Assembly, Logic, and Power Supply, A1.
TA13220E3953	- Printed Wiring Board Assembly, Logic, A2.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongrovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The alarm and status monitor modules (hereinafter called "modules") shall be in accordance with top asemblies TA13220E3950 and TA13220E4050, and as specified herein. The modules are installed in monitor cabinets conforming to MIL-C-52876 as required to display the statuses existing at the protected areas.

3.1.1 Major components. The major components of the modules are as follows:

a. Printed Wiring Board Assembly, Logic, and Power Supply, A1, TA13220E3952 (hereinafter called "PWB A1") which is used in both the alarm and status monitor modules.

b. Printed Wiring Board Assembly, Logic, A2, TA13220E3953 (hereinafter called "PWB A2") which is used only in the status monitor module.

When specified (see 6.2), any of the above major components shall be furnished separately.

3.1.2 Function.

3.1.2.1 Alarm monitor module. The alarm monitor module (hereinafter called "alarm module") monitors and displays the alarm and no-alarm condition of one control unit conforming to MIL-C-52853. The alarm condition is displayed by two redundant alarm lamps. The no-alarm condition is indicated by the extenguishing of the alarm lamps. Each change of status to the alarm state causes the alarm lamps to begin flashing on and off. Activation of an acknowledge switch causes the lamps to go on steady. Subsequent activation of a reset switch extinguishes the lamps if the alarm status at the input has been removed.

3.1.2.2 Status monitor module. The status monitor module (hereinafter called "status module") monitors and displays the status and mode of operation of one control unit conforming to MIL-C-52853. Four different conditions (secure, access, alarm, and alternating current (ac) power) and their inverse are displayed by means of four pairs of status lamps labeled secure, access, ac power, and alarm. The no-alarm and the ac power-fail conditions are indicated by the extinguishing of the alarm and the ac power lamps respectively. Each change of status or mode of operation from one state to the other, except a change from the alarm to no-alarm state, causes the lamps associated with the new status to flash on and off. Activation of an acknowledge switch causes the lamps to go on steady or off corresponding to the status condition then being received. Alarm lamps may only be extinguished by activating a reset switch after the alarm input has been removed.

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3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the modules, or major components when furnished separately. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The modules shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the modules may be newly fabricated from recovered materials to the maximum extent practicable, provided the modules produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the modules.

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3.4 Environmental conditions. The modules and major components when furnished separately shall withstand individual or probable combinations of the environmental conditions specified herein without mechanical or electrical damage (see 6.4.2), performance degradation, or malfunction when tested in accordance with 4.5.2.5.

3.4.1 Operational.

a. Temperature range of -20° F to +150° F.

b. Humidity up to 95 percent relative.

3.4.2 Nonoperational and storage.

a. Temperature range of -30° F to +165° F.

b. Humidity up to 95 percent relative.

c. Shock: 20 times the acceleration due to gravity (g's) of 11 milliseconds (ms) duration and bench handling.

d. Vibration

3.5 Detailed technical requirements.

3.5.1 Monitor unit interface. Connector pin assignments and functions for connector J1 between the monitor cabinet and the modules shall be as follows:

Pin No.	Function
1	Instantaneous alarm/no-alarm input
2	Latched alarm/no-alarm input
3	Access/secure input 1/
4	ac power/ac power fail input 1/
5	Common return for status inputs
6	Alarm output
7	Spare
8	Data receiver transmission line input
9	+20 Volts direct current (Vdc) input
10	Return, direct current (dc)
11	Spare
12	Spare
13	Spare
14	Audible signal buss
15	Spare
16	Data receiver transmission line input

1/ Access/secure and ac power inputs are not utilized by the Alarm Monitor Module.

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3.5.2 Data transmission system interface. Connector pin assignments and functions for connector J2 between the data receiver and modules shall be as follows:

Pin No.	Funtion
1	Instantaneous alarm/no alarm input
2	Latched alarm/no alarm input
3	Access/secure input
4	AC power/ac power fail input
5	Common return for inputs and outputs
6	Spare
7	Spare
8	Data receiver transmission line output
9	+20 Vdc to data receiver
10	Return, dc
11	Spare
12	Spare
13	Spare
14	Spare
15	Spare
16	Data receiver transmission line output

3.5.3 Input status electrical interface. The status inputs presented to the modules shall be impedances which shall be processed as follows:

Pin No.	Impedance to pin 5	State indicated
1, 2	$\geq$ 100 Kilohms (Kohms)	Alarm
1, 2	$\leq$ 2 Kohms	No alarm
3	$\geq$ 100 Kohms 1/	Secure
3	$\leq$ 2 Kohms 1/	Access
4	$\geq$ 100 Kohms 1/	Power fail, ac
4	$\leq$ 2 Kohms 1/	Power on, ac

1/ The alarm module does not respond to impedance changes on terminals 3 and 4.

3.5.4 Power requirements. The maximum regulated +20 Vdc current drain for the status module shall be 100 milliamperes (mA) when four of the incandescent bulbs are illuminated. The maximum current drain for the alarm module shall be 50 mA when the two alarm bulbs are illuminated.

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3.5.5 Operation. Operation of the modules shall be as follows: The high and low impedance states specified in 3.5.3 shall be recognized as the alarm, no alarm, secure, access, ac power fail, and ac power on status conditions of the control unit. The alarm and no alarm conditions will be presented between pins 1 and 5 and between pins 2 and 5. The secure and access conditions will be presented between pins 3 and 5, and the ac power fail and ac power on conditions will be presented between pins 4 and 5 of the input connector. The modules shall indicate an alarm condition when an alarm state (high impedance) is presented at either of the alarm input terminal pairs. With the sonalert circuit connected to the audible signal buss (terminal pin 14) as shown in figure 1 of the appendix, the module operation shall be as follows:

a. On the status module, when a secure condition is received, the secure lamps shall be held on continuously and the access lamps shall be off. When an access condition is being received, the access lamps shall be held on continuously and the secure lamps shall be off. When the access/secure input changes from one status to the other, the lamps corresponding to the new condition shall flash with a period of 250 ±40 milliseconds (ms) with equal on and off times (the flashing lights shall always correspond to the status condition being received), the lamps corresponding to the previously received condition shall be continuously on, and the audible signal shall be activated by the application of a ground potential to the audible signal buss. When the acknowledge switch is momentarily depressed, the flashing lamps shall be changed to continuously on, the lamps corresponding to the previously received condition shall be extinguished and the audible signal shall be silenced by the application of a voltage level between +4 Vdc and +6 Vdc to the audible signal buss.

b. On the status and alarm modules, when an alarm condition is initially received (alarm status change from nonalarm to alarm), the alarm lights shall begin to flash with a period of 250 ±40 ms with equal on and off times, and the audible signal shall be activated by the application of a ground potential to the audible signal buss. The alarm lamps shall continue to flash and the audible signal shall remain activated even if the alarm condition disappears from the modules' input terminal(s). When the acknowledge switch is momentarily activated, the flashing lamps shall be changed to continuously on and the audible signal shall be silenced by the application of a voltage level between +4 Vdc and +6 Vdc to the audible signal buss. The alarm lamps shall remain continuously on and the audible signal shall remain silent even if the alarm condition disappears from the modules' input terminal(s) after the acknowledge switch has been activated. If the alarm condition disappears at any time from the modules' input terminal(s) and later reappears after the acknowledge switch has been activated, then the alarm lamps shall again flash and the

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audible signal shall be activated as specified above. With the alarm lamps on steady, a momentary actuation of the reset switch shall extinguish the alarm lamps if the nonalarm condition is present when the switch is actuated. Actuation of the reset switch shall not extinguish the alarm lamps if the alarm condition is still present. Alarm input conditions shall not affect the access/secure status condition. The condition of the alarm lamps (flashing, continuously on, or off) shall be made available, suitably buffered, at pin 6 of the monitor unit connector. The on condition shall be presented as a voltage level between 0 Vdc and +0.5 Vdc. The off condition shall be presented as a voltage level between +4 Vdc and +6 Vdc. The maximum current requirement in both the on and off condition shall be 2 mA.

c. On the status module, when ac power on condition is being received, the ac power lights shall be on continuously. When the ac power fail condition is being received, the ac power lights shall be off. When a change in ac power status occurs, the ac power lights shall begin to flash with a period of 250 ±40 ms with equal on and off times, and the audible alarm signal shall be actuated by the application of a ground potential to the audible signal buss. When the acknowledge switch is momentarily depressed, the flashing lights shall go to continuously on (if the status at the time of acknowledge is ac power on) or to off (if the status at the time of acknowledge is ac power off) and the audible signal shall be silenced by the application of a voltage level between +4 Vdc and +6 Vdc to the audible signal buss.

3.5.6 Printed wiring board (PWB) assembly, logic, and power supply, A1 (PWB A1). The PWB A1 shall be as specified herein and on top assembly TA13220E3952. The PWB A1 shall have the following electrical characteristics when +20 Vdc is applied to E2, dc return is connected to E7, E13, and E21, and when installed in a test set-up equivalent to that shown in figure 3 of the appendix. Unless otherwise specified, resistances and voltages are referenced to E7.

3.5.6.1 Dc power. Dc power shall be present on the following terminals:

Terminal	Voltage
E1	20 ±0.1 Vdc
E8	5 ±0.5 Vdc
E19	6.2 ±0.6 Vdc
E20	19.3 ±0.4 Vdc

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3.5.6.2 Alarm/Non-Alarm states. Resistances of less than or equal to 2 Kohms from E3 to E7 and from E5 to E7 shall be recognized as the non-alarm input state and shall result in voltages at E4 of 5 ±1 Vdc and E11 of 5 ±0.5 Vdc provided an acknowledge (see 3.5.6.4) and a reset (see 3.5.6.5) have been performed since the last alarm input state. A resistance of greater than or equal to 100 Kohms from either E3 to E7 or E5 to E7, shall be recognized as the alarm input state. A change from the non-alarm to the alarm input state shall cause the voltage at E4 and E11 to fluctuate in an approximate square wave between less than 1 Vdc and 5 ±1 Vdc, with a period of 250 ±40 ms until an acknowledge is performed.

3.5.6.3 Audible signal bus. In the non-alarm input state, a voltage of 5 ±1 Vdc shall exist at E6, provided an acknowledge has been performed since the last change in resistance state. A change from the non-alarm to the alarm input state shall cause the voltage at E6 to become and remain less than 1 Vdc regardless of any further change in input state, provided an acknowledge has not been performed.

3.5.6.4 Acknowledge. An acknowledge shall be initiated by momentarily connecting E9 to E7. Performing an acknowledge shall cause any fluctuating voltages at E4 and E11 to become steady at less than 1 Vdc, regardless of the input state (alarm or non-alarm). It shall also cause the voltage at E6 to become 5 ±1 Vdc. A change from the non-alarm to the alarm input state shall cause the fluctuating voltages described in 3.5.6.2 at E4 and E11, and the voltage at E6 shall become less than 1 Vdc.

3.5.6.5 Reset. A reset is accomplished by momentarily connecting E10 to E7. Performing a reset after an acknowledge shall cause the voltage at E4 and E11 to be 5 ±1 Vdc if the non-alarm input state is present and shall have no effect if the alarm input state is present, or if the voltages at E4 and E11 are fluctuating as described in 3.5.6.2.

3.5.6.6 Lamp test. When E12 is connected to E13, a voltage of less than +1 volt shall be present at terminal E11.

3.5.7 PWB assembly, logic, A2 (PWB A2). The PWB A2 shall be as specified herein and on top assembly TA13220E3953. The PWB A2 shall have the following electrical characteristics when +20 Vdc is applied to terminal E11, +6.2 Vdc is applied to E10, +5 Vdc is applied to E3, and dc returns are connected to E12, and when installed in a test set-up equivalent to that shown in figure 3 of the appendix. Unless otherwise specified, all voltages and resistances are referenced to ground E12.

3.5.7.1 Ac power. A resistance of less than or equal to 2 Kohms from E2 to E12 shall be recognized as the ac power presence input state, and shall result in a voltage of less than 1 Vdc at E5, provided an acknowledge (see 3.5.7.5) has been performed since the last ac power absence input state. A resistance

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greater than or equal to 100 Kohms from E2 to E12 shall be recognized as the ac power absence input state, and shall result in a voltage of 5 ±.5 Vdc at E5, provided an acknowledge has been performed since the last ac power presence input state. A change from one input state to the other shall cause the voltage at E5 to fluctuate in an approximate square wave between less than 1 Vdc and 5 ±.5 Vdc with a period of 250 ±40 ms until an acknowledge is performed.

3.5.7.2 Secure/Access. A resistance of less than or equal to 2 Kohms from E1 to E12 shall be recognized as the access input state, and shall result in a voltage of less than 1 Vdc at E6 and a voltage of 5 ±.5 Vdc at E4, provided an acknowledge has been performed since the last change to the access input state. A resistance of greater than or equal to 100 Kohms from E1 to E12 shall be recognized as the secure input state and shall result in a voltage of less than 1 Vdc at E4 and 5 ±.5 Vdc at E6, provided an acknowledge has been performed since the last change to the secure input state. A change from the secure to access input state shall cause the voltage at E6 to fluctuate between less than 1 Vdc and 5 ±.5 Vdc with a period of 250 ±40 ms and the voltage at E4 to be less than 1 Vdc. A change from the access to secure input state shall cause the voltage at E4 to fluctuate between less than 1 Vdc and 5 ±.5 Vdc with a period of 250 ±40 ms and the voltage at E6 to be less than 1 Vdc.

3.5.7.3 Audible signal. A change of any of the four input states, ac power presence, ac power absence, access, or secure to another input state shall cause the voltage at E9 to be less than 1 Vdc.

3.5.7.4 Acknowledge. An acknowledge shall be accomplished by momentarily capacitively coupling E8 to ground (E12). Performing an acknowledge shall cause any ongoing oscillations at E4, E5, or E6 to cease and the voltages at these terminals to be steady at the values specified in 3.5.7.1 and 3.5.7.2 appropriate for the existing input states. It shall also cause the voltage at E9 to be 5 ±.5 Vdc.

3.5.7.5 Lamp test. A voltage of +1 V or less shall be present at E4, E5, and E6 when E7 is connected to E12.

3.6 Burn-in. Each module and major component when furnished seperately shall be subjected to a 72-hour burn-in to reduce the number of infant mortality failures (see 4.5.2.6).

3.7 Reliability. The modules shall demonstrate a mean-time-between-failures (MTBF) of 10,000 hours when tested as specified in 4.5.2.7.

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3.8 Identification marking. The modules and major components when furnished separately shall be identified in accordance with MIL-STD-130.

3.9 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.5) for manufacturing and testing modules and major components when furnished separately:

Item No.	Description	Identification	Quantity
1.	Stable base drawings of PWB A1	13220E3962, Sh 2 & 3	1 per sheet (See Note a)
2.	Stable base drawing of PWB Interconnect	13220E3963, Sh 2	1 (See Note b)
3.	Stable base drawing of PWB A2	13220E3964, Sh 2 & 3	1 per sheet (See Note c)
4.	Stable base drawing of PWB Conn. left	13220E4055, Sh 2	1 (See Note d)
5.	Stable base drawing of PWB Conn. right	13220E4056, Sh 2	1 (See Note d)
6.	Stable base drawing of PWB Conn. left	13220E4060, Sh 2	1 (See Note c)
7.	Stable base drawing of PWB Conn. right	13220E4061, Sh 2	1 (See Note c)

NOTES:

a. Provided when procuring modules or PWB A1.

b. Provided when procuring status module, PWB A1 or PWB A2.

c. Provided when procuring status module or PWB A2.

d. Provided when procuring alarm module or PWB A1.

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3.10 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.11 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.11.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any test or examination by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Government-loaned property shall be excluded from the requirements of this paragraph.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article modules and major components when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first article modules and major components when furnished separately shall be subjected to the following tests, except that, when specified (see 6.2), the module reliability test may be deleted.

4.3.2.1 Complete modules. The complete module shall be subjected to the tests marked "X" in column 1 of table I. Individual testing of major components of the complete module is not required. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 Major components when furnished separately. Major components when furnished separately shall be subjected to the tests marked "X" in column 1 of tables II and III. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Each module and major component when furnished separately shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual.

4.4.3.1.1 Complete modules. Each complete module shall be subjected to the tests marked "X" in column 2 of table I. Individual testing of major components of the complete module is not required. Failure of any test shall be cause for rejection.

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4.4.3.1.2 Major components when furnished separately. Each major component when furnished separately shall be subjected to the test marked "X" in column 2 of tables II and III. Failure of any test shall be cause for rejection.

4.4.3.2 Samples.

4.4.3.2.1 Complete modules. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. Individual testing of major components of the complete module is not required. AQL shall be 2.5 percent defective.

4.4.3.2.2 Major components when furnished separately. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of tables II and III. AQL shall be 2.5 percent defective.

4.4.4 Noncompliance. If a sample module or major component fails to pass inspection, the manufacturer shall notify the procuring activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the procuring activity has been taken. After the corrective action has been taken, inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the procuring activity). Final acceptance and shipment shall be withheld until the inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the procuring activity.

4.5 Inspection procedure.

4.5.1 Examination. The modules and major components when furnished separately shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Used, rebuilt or remanufactured components, pieces or parts incorporated into the modules.
106. Components missing or not as specified.

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107. Assembly not correct.
108. Identification or special marking missing, incomplete, or illegible.
109. Interchangeability not as specified.
110. Workmanship not as specified.
111. Soldering not as specified.

4.5.2 Tests.

4.5.2.1 Test condition. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. After successful completion of functional testing, each module and major component when furnished separately shall be subjected to a 72-hour burn-in as specified in 4.5.2.6. Unless otherwise specified herein, the modules and major components when furnished separately shall operate as specified herein without maintenance.

4.5.2.2 Test schedule.

TABLE I. Test schedule for complete modules.

First article	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	X	Functional	3.5.5	4.5.2.4
X	X	-	Burn-in	3.6	4.5.2.6
X	-	X	Low temperature	3.4.1, 3.4.2	4.5.2.5.1
X	-	X	High temperature	3.4.1, 3.4.2	4.5.2.5.2
X	-	-	Shock	3.4.2	4.5.2.5.3
X	-	-	Vibration	3.4.2	4.5.2.5.4
X	-	X	Humidity	3.4.1, 3.4.2	4.5.2.5.5
X	-	-	Reliability	3.7	4.5.2.7

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TABLE II. Test schedule for PWB A1.

First article	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	X	Functional	3.5.6	4.5.2.4
X	X	-	Burn-in	3.6	4.5.2.6
X	-	X	Low temperature	3.4.1, 3.4.2	4.5.2.5.1
X	-	X	High temperature	3.4.1, 3.4.2	4.5.2.5.2
X	-	X	Humidity	3.4.1, 3.4.2	4.5.2.5.5

TABLE III. Test schedule for PWB A2.

First article	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	X	Functional	3.5.7	4.5.2.4
X	X	-	Burn-in	3.6	4.5.2.6
X	-	X	Low temperature	3.4.1, 3.4.2	4.5.2.5.1
X	-	X	High temperature	3.4.1, 3.4.2	4.5.2.5.2
X	-	X	Humidity	3.4.1, 3.4.2	4.5.2.5.5

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4.5.2.3 Test methods. Tests shall be conducted in accordance with test methods in the appendix and as specified herein. Calibration of test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Functional. Functional tests of the modules and major components when furnished separately shall be in accordance with 4.5.2.4.1 through 4.5.2.4.4 as applicable.

4.5.2.4.1 Alarm module. The alarm module shall be tested in accordance with appendix test method 1. Nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.4.2 Status module. The status module shall be tested in accordance with appendix test method 2. Nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.4.3 PWB A1. The PWB A1 shall be tested in accordance with appendix test method 3. Nonconformance to 3.5.6 shall constitute failure of this test.

4.5.2.4.4 PWB A2. The PWB A2 shall be tested in accordance with appendix test method 4. Nonconformance to 3.5.7 shall constitute failure of this test.

4.5.2.5 Environmental. The modules and major components when furnished separately shall be subjected to the following series of tests specified in MIL-STD-810. Functional tests required before, during, or after environmental tests shall be in accordance with 4.5.2.4.

4.5.2.5.1 Low temperature. Conduct low temperature test method 502.1, procedure 1, storage temperature of -30° F for not less than 48 hours, equipment operating temperature of -20° F.

4.5.2.5.2 High temperature. Conduct high temperature test method 501.1, procedure II, except that the nonoperating temperature shall be +165° F and the operating temperature shall be +150° F.

4.5.2.5.3 Shock.

a. Basic design: Conduct test method 516.2, procedure I, equipment not operating, sawtooth shock pulse, peak value 20 g's duration 11 ms.

b. Bench handling: Conduct test method 516.2, procedure V.

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4.5.2.5.4 Vibration. Conduct test method 514.2, procedure X, curve AW, equipment not operating.

4.5.2.5.5 Humidity test. Conduct test method 507.1, procedure I with the following exceptions:

a. Step 5 shall be for 5 cycles (not less than 120 hours). Perform measurements at the completion of each cycle.

b. During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.

c. During step 4, the relative humidity shall be maintained at 92 percent or greater.

d. Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle.

4.5.2.5.6 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.5.2.6 Burn-in. Each module and major component when furnished separately shall be subjected to a 72-hour burn-in after successfully completing its functional test and prior to performing further tests. During burn-in, each unit shall be subjected to an abbreviated test as specified herein nine times per 24-hour period, but no test shall be initiated less than 45 minutes after completion of the previous test period. The required tests shall be as follows:

Alarm Module: in successive test periods, alternate between the following two sequences of steps from Appendix Test Method 1, paragraph 3

1) steps f, g, and e in that order.
2) steps m, g, and e in that order.

Status Module: in successive test periods, alternate between the following two sequences of steps from Appendix Test Method 2, paragraph 3:

1) steps f, g, e, q, s, w, y in that order.
2) steps m, g, e, t, v, z, aa in that order.

PWB A1: in successive test periods alternate between performing steps j, k, 1, and m of Appendix Test Method 3, paragraph 3, and performing those same steps except substituting "latched" for "instantaneous" in steps j and k.

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PWB A2: in successive test periods, alternate between the following two sequences of steps from Appendix Test Method 4, paragraph 3:

1) steps b, g, k, l.
2) steps h, i, m, o.

Non-compliance with the applicable portions of 3.6 shall constitute a failure. Failures during burn-in shall be analyzed, repaired and corrective action taken as necessary. Copies of failure reports (see 6.2) shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional 72-hour burn-in, after failure analysis and repair, and successful completion of the functional test.

4.5.2.7 Reliability. Using an upper test MTBF (θ_0) of 10,000 hours and a lower test MTBF (θ_1) of 5000 hours, the modules shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. A reliability failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.

b. Serious damage to the system.

c. Serious personnel safety hazards.

4.5.2.7.1 Test procedure. Tests shall be conducted under ambient laboratory conditions, without temperature or humidity cycling. The equipment shall be energized at the normal input voltage of 20 $\pm$1 Vdc. Operation shall be continuous throughout the test with all circuits exercised through all functions not less than nine times during each 24-hour period and not more than once per hour. The parameters to be measured shall be in accordance with the applicable appendix test method. Reliability tests shall commence after installation and a burn-in period of 72 hours. Failures of indicator lamps are not relevant, but any dependent failure caused thereby are relevant. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified if practicable. This accounting shall be maintained separately from the failure accounting. Redundant elements, such as indicator lamps, shall not be repaired nor counted as a failure until the status module or alarm module has failed to perform its intended function. Preventive maintenance procedures specified for the equipment during normal operation shall be performed during the reliability tests. No additional preventive maintenance is allowed during the test.

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4.5.2.7.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112.	Preservation methods for modules and major components not as specified for levels A or B.
113.	Boxed contents not cushioned to prevent movement or damage as specified for level A or level B.
114.	Consolidation not as specified for level A or level B.
115.	Strapping not as specified for level A.
116.	Marking missing, illegible, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. The module components shall be preserved by one of the following procedures, depending upon procurement of complete modules or procurement of major components when furnished separately.

5.1.1.1 Preservation of complete modules. Each status and alarm module shall be preserved in accordance with MIL-P-116, method IIc. Multiple modules of the same type shall then be placed together in a close-fitting box conforming to PPP-B-636, W6c or W5c, style optional. The components shall be cushioned within the box to prevent movement or damage.

5.1.1.2 Major components when furnished separately. Major components, when procured separately, shall be individually preserved in accordance with MIL-P-116, Method IIc. Each component shall then be placed in a close-fitting box conforming to PPP-B-566, variety 1, style, type, and class optional or PPP-B-676, variety 1, style, type, and class optional. The components shall be cushioned to prevent movement or damage within the box.

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5.1.1.2.1 Consolidation. Components of like description, preserved as specified in 5.1.1.2, shall be placed together in a close-fitting box conforming to PPP-B-636, W5c, style optional, in quantities not to exceed the weight or size limitations of the box.

5.1.2 Level B. The module components shall be preserved as specified in 5.1.1 for level A, except the method of preservation shall be method IA-8.

5.1.3 Commercial. Each module and major component when procured separately, shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The module components, preserved as specified in 5.1, shall be packed by one of the following procedures depending upon procurement of complete modules or procurement of major components furnished separately.

5.2.1.1 Packing of complete modules. Complete modules shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed the weight limitations of the box. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.1.2 Packing of major components when furnished separately. Major components of like description shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Complete modules and major components when furnished separately and preserved as specified in 5.1 shall be packed as specified in 5.2.1.1 or 5.2.1.2 as applicable, except boxes shall be Domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitations of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Complete modules and major components furnished separately and preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for level A or level B shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

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5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951 with the added requirement for weight and cube data on the shipping container. Special marking specified in 5.3.3 shall also be included.

5.3.3 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the project code "JID".

6. NOTES

6.1 Intended use. The modules are intended for use as components of the Joint-Services Interior Intrusion Detection System (J-SIIDS), which is used to protect indoor areas at Government installations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and quantity of modules and major components required (see 3.1 and 3.1.1).

c. When a first article is required for inspection and approval, the number of units required, and allocation of units to environmental and reliability tests respectively (see 3.2 and 6.3).

d. When Government-loaned property is not required (see 3.9).

e. Whether first article reliability testing may be deleted (see 4.3.2).

f. Degree of preservation and packing required (see 5.1 and 5.2).

g. When other than finish B strapping is required (see 5.2.1.1 and 5.2.1.2).

h. Schedule, content and format of burn-in failure reports (see 4.5.2.6).

6.3 First article. When a first article inspection is required, the item(s) should be an initial production item. The first article should consist of one or more unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the status module or alarm module is designed to alarm.

6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any component; physical change which impairs the mechanical integrity of the status module or alarm module; evidence of delamination or water penetration into integrated circuits, PWBs or external components or physical conditions resulting in noncompliance of the modules with the provisions of this specification.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.9, when applicable (see 6.2).

Custodians:
Army - ME
Navy - EC
Air Force - 99

Review Activity:
DLA - GS
Air Force - 82
Navy - MC

User activities:
Navy - CG, SH

Preparing activity:
Army - ME

Project 6350-0099

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APPENDIX

TEST METHODS

FOR

MONITOR MODULE, STATUS, ID-1921/FSS-9(V)

AND

MONITOR MODULE, ALARM, ID-1922/FSS-9(V)

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether modules and major components, when furnished seperately, procured under this specification conform to the requirements specified. Figures 1 through 3 contain data for fabrication of test fixtures which are required to facilitate testing of the modules and major components, when furnished seperately.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX

TEST METHOD 1

ALARM MODULE FUNCTIONAL TEST

1. TEST APPARATUS:

   a. Test set - See figure 1.
   b. Power supply, dc - Preston Scientific X-MOD-101 or equivalent.
   c. Oscilloscope - Tektronix 305 or equivalent.
   d. Multimeter - Simpson 260 or equivalent.

2. TEST PREPARATION:

   a. Connect test equipment and test fixture as shown in figure 1. Set switches S1 and S2 to their closed positions.
   b. Turn on the dc power supply and set it for 20 Vdc output. Turn the power supply off and connect it and the multimeter to the + 20 Vdc input (TB1) of the test set as shown in figure 1.
   c. Remove the right-hand cover of the alarm module.
   d. Connect J1 of the test set to plug J1 of the alarm module.
   e. Set the multimeter to the 100 ma scale.
   f. Connect a ground lead from the oscilloscope to TB1-2 on the test set. Attach an X10 probe to input 1 of the oscilloscope. Set the oscilloscope to channel 1 on the vertical selector switch. Set the triggering controls to channel 1, dc, positive slope, and automatic. Set the channel attenuator to 0.1 volt per division and the time base to 0.1 sec/div. Turn the oscilloscope on.
   g. Turn the power supply on. The sonalert shall be silent and the alarm lamps shall be off.

3. TEST PROCEDURE:

   a. Open switch S1. The sonalert shall sound and the two alarm lamps shall flash.
   b. Touch the X10 probe of the oscilloscope to the terminal lug of one of the alarm lamps. A square wave shall be observed on the oscilloscope with a low-level voltage between 0 Vdc and + 0.5 Vdc, and a high-level voltage of 5 $\pm$ 0.5 Vdc, and a period of 0.25 $\pm$ 0.04 seconds. The period is the time between two consecutive positive or negative transitions.
   c. Close switch S1. The sonalert shall continue to sound and the alarm lamps to flash.

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d. Momentarily place the acknowledge/reset switch in the ACK position. This accomplishes an acknowledge. The sonalert shall stop sounding and the two alarm lamps shall stop flashing and turn on continuously.
e. Momentarily place the acknowledge/reset switch in the RESET position. This accomplishes a reset. The alarm lamps shall turn off.
f. Open switch S1 momentarily. The sonalert shall sound and the two alarm lamps shall flash.
g. Accomplish an acknowledge. The sonalert shall stop sounding and the two alarm lamps shall turn on steady.
h. Observe the current reading on the multimeter. It shall not exceed 50 ma.
i. Open switch S1 momentarily. The sonalert shall sound and the alarm lamps shall flash.
j. Momentarily place the acknowledge/reset switch in the RESET position. The sonalert shall continue to sound and the alarm lamps to flash.
k. Accomplish an acknowledge and a reset. The sonalert and the alarm lamps shall turn off.
l. Connect the oscilloscope probe to TP1 on the test set. The dc voltage level, as read on the oscilloscope, shall be between + 4 Vdc and + 6 Vdc.
m. Open switch S2 momentarily. The sonalert shall sound and the two alarm lamps shall flash. A square wave shall be observed on the oscilloscope with a low-level voltage between 0 Vdc and + 0.5 Vdc, and a high-level voltage between + 4 Vdc and + 6 Vdc and a period of 0.25 $\pm$ 0.04 seconds.
n. Accomplish an acknowledge. The alarm lamps shall be on steady and the audible signal shall be silenced. The dc voltage level, as read on the oscilloscope, shall be between 0 Vdc and 0.5 Vdc.
o. Accomplish a reset. The alarm lamps shall turn off and the voltage observed on the oscilloscope shall return to between + 4 and + 6 Vdc.
p. Momentarily place the lamp test switch in the TEST position. The alarm lamps on the alarm module shall turn on while the switch is in the TEST position. The lamps shall turn off as soon as the switch is removed from the TEST position.
q. Perform continuity checks to confirm the following connections between terminals on J1 and J2 of the alarm module: J1-1 to J2-1; J1-2 to J2-2; J1-5 to J2-5; J1-8 to J2-8; J1-9 to J2-9; J1-10 to J2-10; J1-16 to J2-16; J1-5 to J1-10.
r. This concludes the functional test for the alarm module.

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APPENDIX

TEST METHOD 2

STATUS MODULE FUNCTIONAL TEST

1. TEST APPARATUS:

   a. Test set - See figure 1.
   b. Power supply, dc - Preston Scientific X-MOD-101 or equivalent.
   c. Oscilloscope - Tektronix 305 or equivalent.
   d. Multimeter - Simpson 260 or equivalent.

2. TEST PREPARATION:

   a. Connect test equipment and test fixture as shown in figure 1. Set switches S1 through S4 to their closed positions.
   b. Turn on the dc power supply and set it for 20 Vdc output. Turn the power supply off and connect it and the multimeter to the + 20 Vdc input (TB1) of the test set as shown in figure 2.
   c. Remove the right-hand cover of the status module.
   d. Connect J1 of the test set to plug J1 of the status module.
   e. Set the multimeter to the 500 ma scale.
   f. Connect a ground lead from the oscilloscope to TB1-2 on the test set. Attach an X10 probe to input 1 of the oscilloscope. Set the oscilloscope to channel 1 on the vertical selector switch. Set the triggering controls to channel 1, dc, positive slope, and automatic. Set the channel attenuator to 0.1 volt per division and the time base to 0.1 sec/div. Turn the oscilloscope on.
   g. Turn the power supply on. The sonalert shall be silent and the alarm lamps shall be off.

3. TEST PROCEDURE:

   a. Open switch S1. The sonalert shall sound and the two alarm lamps shall flash.
   b. Touch the X10 probe of the oscilloscope to the terminal lug of one of the alarm lamps. A square wave shall be observed on the oscilloscope with a low-level voltage between 0 Vdc and + 0.5 Vdc, and a high-level voltage of 5 $\pm$ 0.5 Vdc, and a period of 0.25 $\pm$ 0.04 seconds. The period is the time between two consecutive positive or negative transitions.

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c. Close switch S1. The sonalert shall continue to sound and the alarm lamps to flash.
d. Momentarily place the acknowledge/reset switch in the ACK position. This accomplishes an acknowledge. The sonalert shall stop sounding and the two alarm lamps shall stop flashing and turn on continuously.
e. Momentarily place the acknowledge/reset switch in the RESET position. This accomplishes a reset. The alarm lamps shall turn off.
f. Open switch S1 momentarily. The sonalert shall sound and the two alarm lamps shall flash.
g. Accomplish an acknowledge. The sonalert shall stop sounding and the two alarm lamps shall turn on steady.
h. Open switch S1 momentarily. The sonalert shall sound and the alarm lamps shall flash.
i. Momentarily place the acknowledge/reset switch in the RESET position. The sonalert shall continue to sound and the alarm lamps to flash.
j. Accomplish an acknowledge and a reset. The sonalert and the alarm lamps shall turn off.
k. Connect the oscilloscope probe to TP1 on the test set. The dc voltage level, as read on the oscilloscope, shall be between + 4 Vdc and + 6 Vdc.
l. Open switch S2 momentarily. The sonalert shall sound and the two alarm lamps shall flash. A square wave shall be observed on the oscilloscope with a low-level voltage between 0 Vdc and + 0.5 Vdc, and a high-level voltage between + 4 Vdc and + 6 Vdc and a period of 0.25 $\pm$ 0.04 seconds.
m. Accomplish an acknowledge. The alarm lamps shall be on steady and the audible signal shall be silenced. The dc voltage level, as read on the oscilloscope, shall be between 0 Vdc and 0.5 Vdc.
n. Accomplish a reset. The alarm lamps shall turn off and the voltage oscerved on the oscilloscope shall return to between + 4 and + 6 Vdc.
o. Open switch S3. The sonalert shall sound, the access lamps shall be on steady, and the secure lamps shall be flashing.
p. Repeat step b., except apply probe to terminal lug of one of the secure lamps.
q. Accomplish an acknowledge. The access lamps shall go off, the secure lamps shall come on steady, and the sonalert shall be silenced.
r. Close switch S3. The sonalert shall sound, the secure lamps shall be on steady, and the access lamps shall be flashing.
s. Repeat step b., except apply probe to terminal lug of one of the access lamps.
t. Accomplish an acknowledge. The secure lamps shall go off, the access lamps shall come on steady, and the sonalert shall be silenced.
u. Open switch S4. the sonalert shall sound and the ac power lamps shall flash.

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APPENDIX

v. Repeat step b., except apply probe to terminal lug of one of the ac power lamps.
w. Accomplish an acknowledge. The ac power lamps shall go off and the sonalert shall be silenced.
x. Close switch S4. The sonalert shall sound and the ac power lamps shall flash.
y. Accomplish an acknowledge. The ac power lamps shall come on steady and the sonalert shall be silenced.
z. Observe the multimeter. The maximum current as indicated on the meter shall not exceed 100 ma with 2 sets of lamps illuminated.
aa. Momentarily place the lamp test switch in the TEST position. All lamps on the status module not already on shall turn on while the switch is in the TEST position. The lamps shall turn off as soon as the switch is removed from the TEST positon.
ab. Perform continuity checks to confirm the following connections between terminals on J1 and J2 of the status module: J1-1 to J2-1; J1-2 to J2-2; J1-3 to J2-3; J1-4 to J2-4; J1-5 to J2-5; J1-8 to J2-8; J1-9 to J2-9; J1-10 to J2-10; J1-16 to J2-16; J1-5 to J1-10.
ac. This concludes the functional test for the status module.

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APPENDIX

TEST METHOD 3

FUNCTIONAL TEST

PWB A1

(13220E3952)

1. TEST APPARATUS:

   a. Test fixture - See figure 2.
   b. Power supply, dc - Hewlett-Packard 6101A or equivalent.
   c. Oscilloscope - Tektronix 305 or equivalent.
   d. Voltmeter, digital (DVM) - Fairchild 7000A or equivalent.

2. TEST PREPARATION:

   a. Connect test apparatus to test fixture in accordance with figure 2.
   b. Adjust power supply for 20 ±0.1 Vdc output and then turn off power supply.
   c. Attach an X10 probe to input 1 of the oscilloscope. Set the oscilloscope to channel 1 on the vertical selector switch. Set the triggering controls to channel 1, ac, positive slope, and automatic. Set the channel attenuator to 0.1 V per division and the time base to 0.5 second per division. Turn the oscilloscope on.
   d. Turn the instantaneous alarm and latched alarm switches to the NO ALARM position (closed).
   e. Connect pwb to test fixture plugs P1, P2, and P3.

3. TEST PROCEDURE:

   a. Turn on the dc power supply. The alarm lamps shall be off.
   b. Select the appropriate scale on the DVM and rotate monitor switch S4 as required to measure voltage levels at the following pins on PWB A1. Voltages shall be as follows:

E8--	5 ± 0.5 Vdc
E1--	20 ±0.1 Vdc (same as E2, 20 V input)
E19--	6.2 ±0.6 Vdc (VDD).
E20--	19.3 ±0.4 Vdc

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c. Actuate the lamp test switch. The alarm lamps shall be on steady while the switch is actuated.
d. Rotate S4 to monitor pin E4, the buffered alarm output, with the DVM. The DVM shall indicate between 4 and 6 V.
e. Turn the instantaneous alarm switch on the test set to the ALARM position. The sonalert shall sound and the alarm lamps shall be flashing.
f. Observe the oscilloscope. It shall display a square wave with a low level of less than 1 V and a high level of 5 $\pm$ 1 V.
g. Momentarily depress acknowledge switch, S3. The sonalert shall be silenced and the alarm lamps shall be on steady.
h. Momentarily depress reset switch. The sonalert shall remain silent and the alarm lamps shall remain on steady.
i. Turn the instantaneous alarm switch to the NO ALARM position. The alarm lamp shall remain on steady.
j. Turn the instantaneous alarm switch to the alarm position. The sonalert shall sound and the alarm lamps shall be flashing.
k. Turn the instantaneous alarm switch to the NO ALARM position. The sonalert shall continue to sound and the alarm lamps to flash.
l. Momentarily depress acknowledge switch. The sonalert shall be silenced and the alarm lamps shall be on steady.
m. Momentarily depress reset switch. The alarm lamps shall be off.
n. Repeat steps e. through m. and substitute latched alarm for instantaneous alarm for all cases where instantaneous alarm is specified.
o. Turn the latched alarm switch to the ALARM position momentarily. The sonalert shall sound and the alarm lamps shall be flashing.
p. Rotate monitor switch S4 to monitor E11. The oscilloscope shall display a square wave with a low level of less than 1 V and a high level of 5.0 $\pm$ 1.0 V and a period of 0.25 $\pm$ .04 second.
q. Actuate the acknowledge and reset switches. The sonalert and lamps shall be off.
r. Turn off the dc power supply. This completes the functional test.

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APPENDIX

TEST METHOD 4

FUNCTIONAL TEST

PWB A2

(13220E3953)

1. TEST APPARATUS:

   a. Test fixture - See figure 3.
   b. Power supply, dc - Hewlett-Packard 6101A or equivalent.
   c. Oscilloscope - Tektronix 305 or equivalent.
   d. Voltmeter, digital (DVM) - Fairchild 7000A or equivalent.

2. TEST PREPARATION:

   a. Connect test apparatus to test fixture in accordance with figure 3.
   b. Apply 20 ±0.1 V to terminals labelled "V in" in figure 3.
   c. Adjust R12 and R14 to obtain voltages at E10 and E3 of 6.2 ±0.1 V and 5.0 ±0.1 volts respectively.
   d. Temporarily tie E9 to ground as required to adjust R7 to obtain a suitable sonalert output level. Turn off power supply after adjusting.
   e. Place S1 and S2 in their closed positions.
   f. Attach an X10 probe to input 1 of the oscilloscope. Set the oscilloscope to channel 1 on the vertical selector switch. Set the channel 1 attenuator to 0.1 V per division and the time base to 0.5 second per division. Turn the oscilloscope on. Attach the probe to wiper of switch S5 as shown in figure 3.
   g. Connect pwb under test to test fixture connectors P1, P2, and P3.

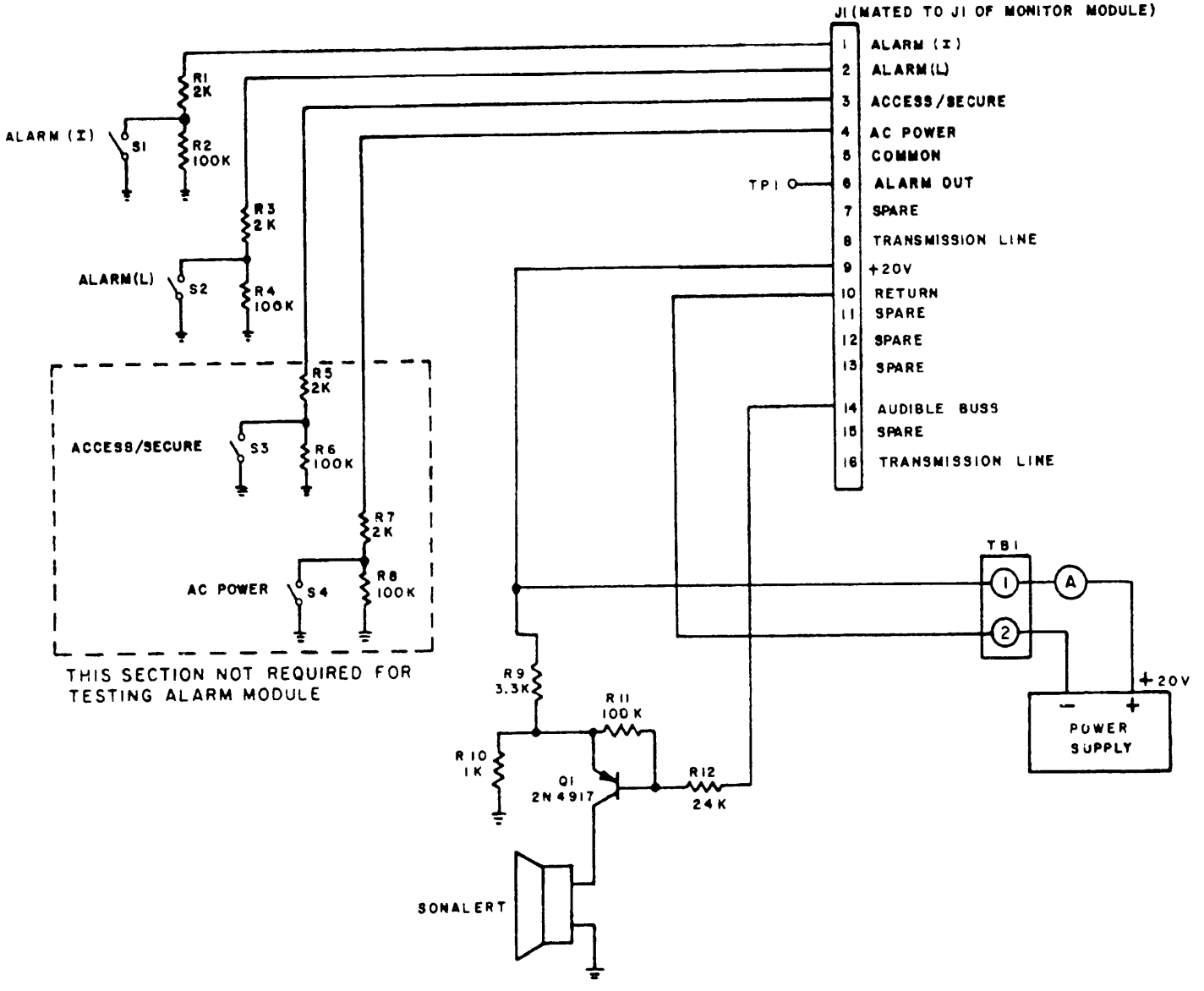
3. TEST PROCEDURE:

   a. Turn on 20 Vdc power supply. The access lamps shall be on steady and the secure lamps off. The ac power lamps shall be on steady and shall remain so through step j.
   b. Open switch S1. The sonalert shall sound, the access lamps shall be on steady and the secure lamps shall be flashing.
   c. Set monitor switch S5 to SEC. A square wave of amplitude 5 ± 0.5 V p-p and period (time between consecutive positive transitions) 0.25 ± 0.04 second shall be displayed on the oscilloscope.
   d. Close switch S1. The sonalert shall continue to sound, the secure lamps shall be on steady, and the access lamps shall be flashing.

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e. Set S5 to ACC. The oscilloscope shall display a square wave of amplitude 5 ± 0.5 V p-p and period 0.25 ± .04 second.
f. Open switch S1. the sonalert shall continue to sound, the access lamps shall be on steady, and the secure lamps shall be flashing.
g. Momentarily depress S4 to perform an acknowledge. The access lamps shall go off, the secure lamps shall come on steady, and the sonalert shall be silenced.
h. Close switch S1. The sonalert shall sound, the secure lamps shall be on steady and the access lamps shall be flashing.
i. Accomplish an acknowledge. The secure lamps shall go off, the access lamps shall come on steady, and the sonalert shall be silenced.
j. Actuate the lamp test switch. The secure lamps shall be on steady while the switch is actuated. The access lamps and ac power lamps shall remain on steady.
k. Open switch S2 (ac power switch).
l. The sonalert shall sound and the ac power lamps shall flash. Accomplish an acknowledge. The ac power lamps shall go off and the sonalert shall be silenced.
m. Close switch S2. The sonalert shall sound and the ac power lamps shall flash.
n. Set monitor switch S5 to ac power. The oscilloscope shall display a square wave of amplitude 5 ± 0.5 V p-p and period 0.25 ± .04 second.
o. Accomplish an acknowledge. The ac power lamps shall come on steady and the sonalert shall be silenced.
p. Turn off the power supply. This completes the functional test for the A2 PWB.

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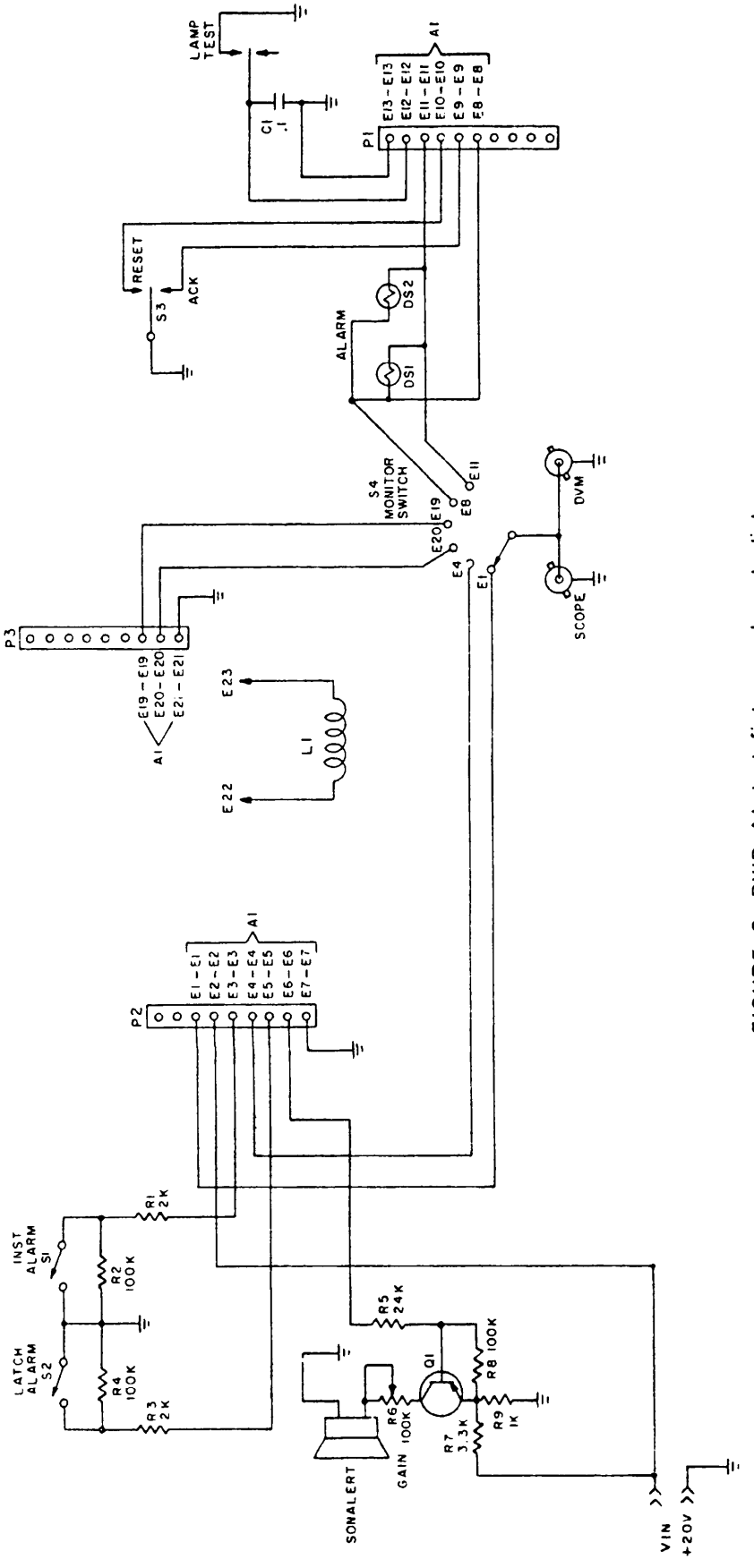
NOTES:

1. RESISTORS SHALL BE CARBON COMPOSITION OR METAL FILM, 5% TOLERANCE OR LESS.
2. SONALERT SHALL BE MALLORY MODEL SC628-72206 OR EQUIVALENT.

FIGURE 1. Alarm and status module test fixture schematic diagram.

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X-4437

FIGURE 2. PWB A1 test fixture and parts list.

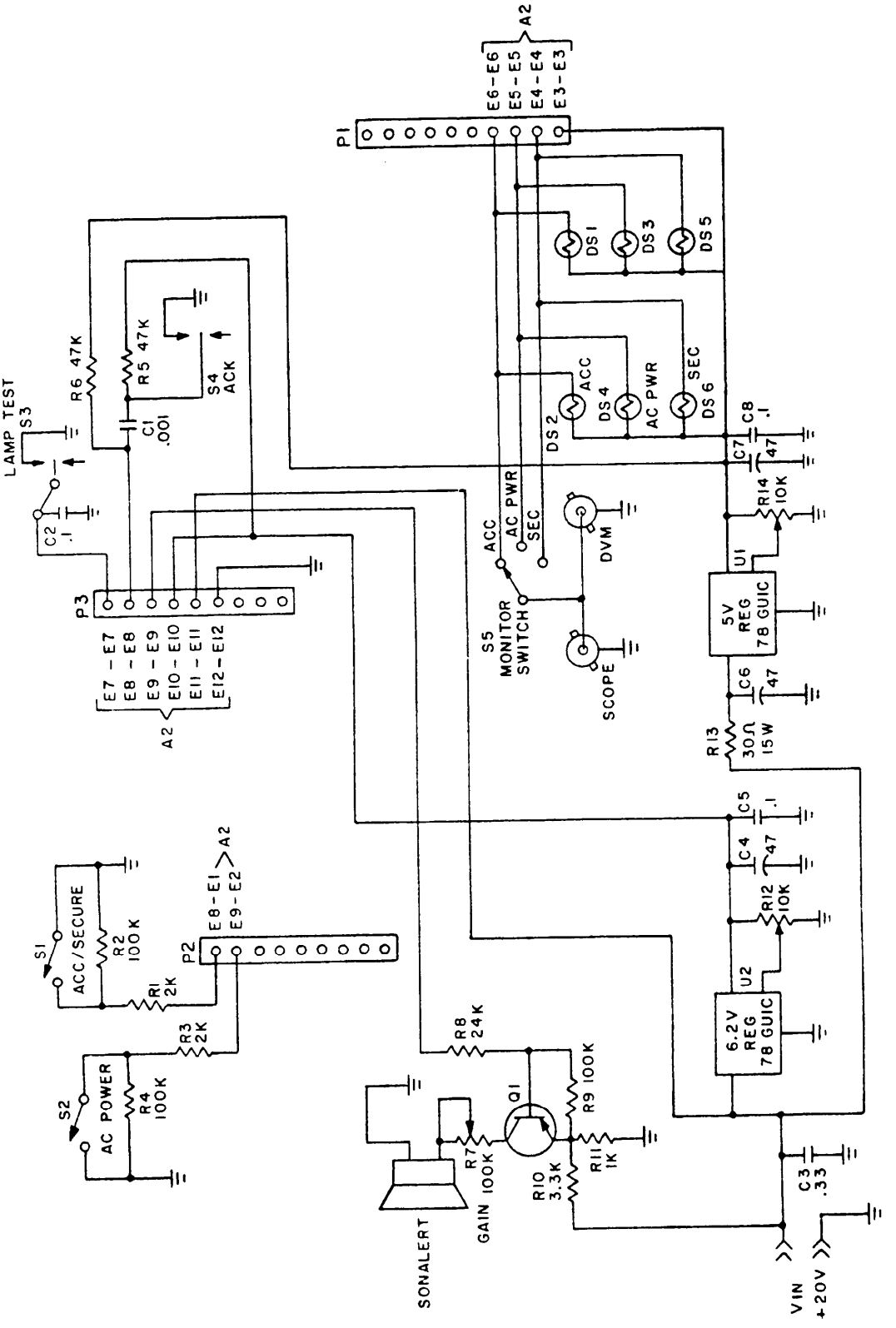
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Resistors	Designation
1K 1/4W 5%	R9
2K 1/4W 5%	R1, R3
3.3K 1/4W 5%	R7
24K 1/4W 5%	R5
100K 1/4W 5%	R2, R4, R6, R8
Capacitor	
.1μf	C1
Switches	
12 Position, 1 Pole, Rotary (CTS T205 or equiv.)	S4 (Monitor)
SPDT Toggle (Dialight 571-1121-0101-011 or equiv.)	S1, S2
SPDT Momentary (ALCO MST-105G or equiv.)	S3 (ACK/Reset) S5 (Lamp Test)
Resistors, Variable	
100K	R7
Transistors	
2N4917	Q1
Transducers	
Sonalert (Mallory SC628-72206)	Sonalert
Lamps	
MS24515-685	DS1, DS2

FIGURE 2. PWB A1 test fixture and parts list. - Continued

X-4437

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FIGURE 3. PWB A2 test fixture and parts list.

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Resistors	Designation
1K 1/4W 5%	R11
2K 1/4W 5%	R1, R3
3.3K 1/4W 5%	R10
24K 1/4W 5%	R8
47K 1/4W 5%	R5, R6
100K 1/4W 5%	R2, R4, R9
30 Ω 15W 10%	R13
Capacitors	
.001µf	C1
.1µf	C2, C5, C8
.33µf	C3
47µf	C4, C6, C7
Switches	
SPDT Toggle (Dialight 571-1121-0101-011 or equiv.)	S1, S2
SPDT Momentary (ALCO MST-105G or equiv.)	S4 (ACK/Reset), S3 (Lamp Test)
3 Position, 1 Pole, Rotary (Centralab 1461 or equiv.)	S5 (Monitor)
Resistors, Variable	
10K	R12, R14
100K	R7
Transistors	
2N4917	Q1
Transducers	
Sonalert (Mallory SC628-72206)	Sonalert
Integrated Circuits	
Voltage regulator (Fairchild µA 78GUIC or equiv.)	U1, U2
Heat Sink	
(Thermalloy 6109PB or equiv.)	U1
Lamps	
MS24515-685	DS1 thru DS8

FIGURE 3. PWB A2 test fixture and parts list. - Continued

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
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